

This is the Swish Analytics Data Engineering Interview Project. There are 2 parts to this project. The first part you will programmatically solve a common data problem leveraging your python skills and sports knowledge. Part 2 focuses on the system implementation of the code you wrote in part 1. Please deliver the necessary code, diagrams, and write ups for both parts in a zip.

Part 1

In the NBA, boxscore data provides a useful but limited view of what happened during the game. It provides all the most useful stats for players who were directly involved; however, it's difficult to evaluate the effect of players that are not directly involved in the outcome of a play, and in order to do so, you first need to know which players were on the court for each play. Your task is to use the provided play-by-play (pbp) data, along with your technical skill and sports knowledge, to solve this problem programmatically.

Goal

Write a Python script which will utilize the provided data sets to create a new data set that details which 10 players were on the court for each play of the game.

Provided Data Sets

- pbp: General play-by-play.
- pbp_players: play-by-play with player-specific events broken out.
- rosters: team rosters for corresponding NBA games.

Requirements

- Script written in Python
- Each play_id from the provided pbp data set must be represented in this new data set.
- For each play_id, there must be 10 players selected as being on court.
- The script writes this new data set to a MySQL table called pbp_players_on_court.
- When the script is run a second time, it updates the new table to catch any new changes.
- Be able to run the script for an individual game, or for all of them, determined by an argument passed to the script.

Tips

- Design this new data set with the assumption that it will be utilized for data analysis, such as summing up the total team points scored while each player was on court. If you want to make sure it's designed intelligently for such a purpose, try doing your own calculations on the new data set.
- While the 3 provided data sets should be the only ones that the script accesses, feel free to use external sources to spot check, cross-reference, or gain any more info you may need about NBA rules.

Deliverables: In a .zip file, please provide the following:

- The Python script. Remove any private connection parameters.

- A MySQL export of the new table, including the CREATE statement and the data your code should generate.
- A written walkthrough of your code. Be sure to explain why you chose the structure you did, identify any shortcomings of your code, and analyze the time complexity of your code. Feel free to include anything that you think would be beneficial for us to know.

Part 2: Now that you've completed the NBA "who's on the court" problem, design a system around your project as if it were in production. Assume all you have is the code you've written and a play-by-play REST API from a data provider.

Use your knowledge in system design to develop a **system diagram** (draw.io or similar) and **write up** to explain your decisions, why you've decided to work with particular tools or technology, and how your system might scale. Your system should be end-to-end, from retrieving the data to making it available for others. Think about how to ingest the play-by-play data, how does your code know when to run, and the storage strategy.

Here's a very basic example diagram:

